FX-JEPA

Learning Effect-Aware Audio Representations for Reference-Based Audio Effect Transfer

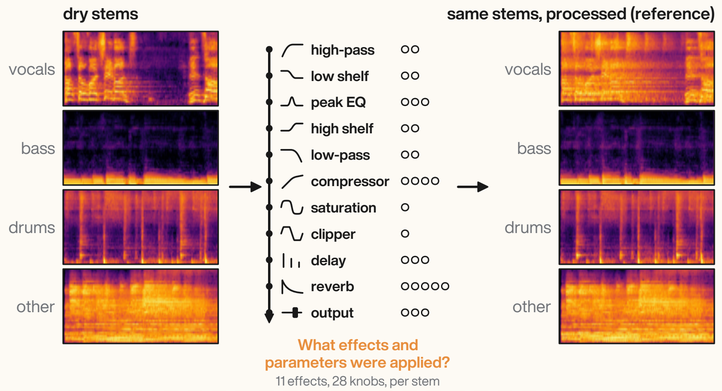
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Universal Audio | SANE 2026, MIT, Oct 30th | Poster AM-10 | mclemens@uaudio.com

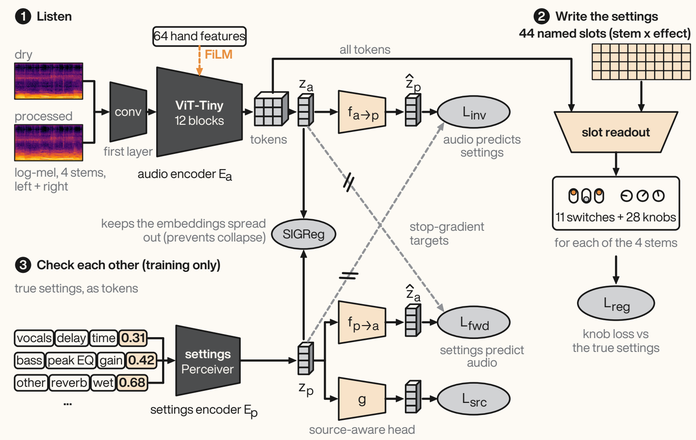
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Reference-based effect style transfer

FX-JEPA predicts the strip settings from the audio embedding, then renders once.

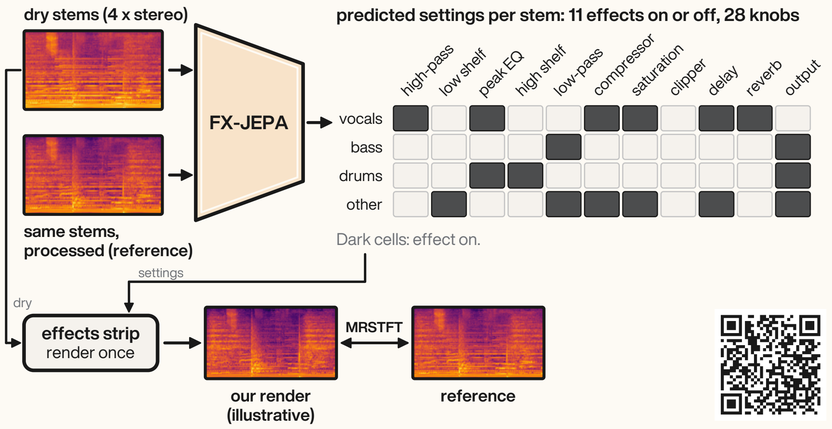


The FX-JEPA model



$$\mathcal{L} = \frac{1-2\lambda}{3}(\mathcal{L}_{inv} + \mathcal{L}_{fwd} + \mathcal{L}_{src}) + \lambda(S_a + S_p) + \mathcal{L}_{reg} + 0.5(\mathcal{L}_{adv} + \mathcal{L}_D + \mathcal{L}_{st})$$

$\lambda = 0.02$. $\mathcal{L}_{inv}$: audio predicts settings. $\mathcal{L}_{fwd}$: settings predict audio. $\mathcal{L}_{src}$: settings plus dry stems predict the pair for each song. S_a, S_p: SIGReg per branch. $\mathcal{L}_{reg}$: the knob loss. $\mathcal{L}_{adv}, \mathcal{L}_D, \mathcal{L}_{st}$: settings side only; none reach the audio encoder.



Audio examples
dry, reference, ours; 8 clips

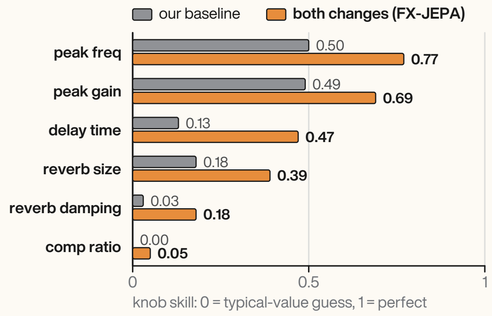
65%
less spectral distance than leaving the stems dry

0.3 s
to predict one clip's settings, on one GPU

10.9%
less spectral distance than our baseline, from the slot readout alone

Ablations

The baseline predicts all 44 settings from one learned summary token. FX-JEPA gives each stem and effect its own learned query over the audio tokens. That change alone cuts the spectral distance by 10.9%, encoder unchanged.



median error, baseline to best	
peak freq	0.48 to 0.14 octaves
peak gain	2.0 to 1.0 dB
delay time	93 to 40 ms
reverb size	decay time 27% to 18% off
reverb damping	treble decay 6% to 5% off
comp ratio	at the guessing level

Spectral distance to the reference

MRSTFT, 1,000 held-out clips from 929 songs; lower is better.

No processing
the dry stems, unchanged
1.17

Best published encoder, as released
best of four: StemFX, AFx-Rep, FX-Encoder++, RelFX; frozen, small readout on top
0.869

StemFX design, trained from scratch on our data
our data and recipe
0.485

FX-Encoder++ design, trained from scratch on our data
our data and recipe; 70k of 108k steps, one run
0.463

Our baseline
ViT-Tiny; all settings from one learned summary token; second run 0.504
0.492

Baseline + conv first layer
convolutions before the ViT; alone, within run-to-run noise
0.495

Baseline + slot readout
one query per stem and effect
0.439

Both changes (FX-JEPA)
conv first layer + slot readout; second run 0.443
0.413

0 0.5 **1.17**

MRSTFT, 0 = identical to the reference; lower is better

Terms

slot readout
one learned query per stem and effect, 44 in all; each reads the audio tokens and predicts that effect's settings

conv first layer
a small convolution stack in place of the ViT's patch embedding, at the input

JEPA
joint embedding predictive architecture; audio and settings embeddings predict each other

knob loss
cross-entropy on the switches and squared error on the knob values, against the true settings

data FMA stems (SCNet separations), MoisesDB stems; public data only; 44.1 kHz, 10 s clips

effects re-implemented from open-source parts: MultiAFx EQ, torchcomp compressor, tanh saturation, hard clipper, feedback delay, Freeverb, output gain, pan and width; fixed order